

Feed By-Products

INTRODUCTION

By-product feeds are defined as “secondary products produced in addition to the principal product” (AAFCO, 2016). These secondary products originate from a wide range of industries, including the food, fiber, beverage, and bioenergy industries. By-products usually originate from production processes where at least a portion of the nutrients from the raw input is removed, the extent of which varies both within and by production process. By-products often represent cost-effective sources of protein and energy and may even improve palatability of many rations (Van Soest, 1994). By-products are often considered inedible by humans, but when fed to a dairy cow, they are converted to high-quality human food. The nutrient and chemical composition of by-products can vary depending on manufacturer, geographical region, or site of origin and often changes over time. Production processes such as excessive heating during drying can reduce the nutritional value and overall quality of by-products.

Beyond their nutritional value, feeding by-products to livestock is advantageous for several reasons. First, most cannot be consumed by humans, and consequently, the use of by-product feeds increases the overall efficiency of human-consumable inputs by the dairy industry. Second, by-products reduce the amount of grain used by livestock, resulting in increased grain available for human consumption (Karlsson et al., 2018). Third, the feeding of by-products to livestock eliminates the need for waste disposal from a variety of industries (Bampidis and Robinson, 2006). Fourth, the production of by-products may even represent a safer feed for cattle. Such is the case with sugar beets, which, if fed, are more likely to cause ruminal acidosis than the fibrous but highly digestible by-product, beet pulp (Crawshaw, 2004). Furthermore, many of these feeds originate from human food production, which follows higher standards than that of feed production. The inclusion of by-products into diets of dairy cattle may be limited by a number of nutritional, technical, and socioeconomic aspects.

The objective of this chapter is to identify major feed by-products used in the dairy industry and to provide clarity for the origin and nomenclature of feeds that are listed, with the chemical composition described in Chapter 19, while also noting any limitations or challenges associated with their use. Where applicable, published reviews are emphasized. Many published studies have been designed to include one or more by-products and replace forages or common commodities such as corn or soybean meal. In some cases, the inclusion rates are high, and often such experiments illustrate the adaptability of the modern dairy cow to produce a high-quality food product. This chapter will summarize key studies but will stop short of recommending so-called optimal inclusion rates as such justification is multifactorial in nature. Furthermore, safe and effective use of these products should follow general feeding recommendations outlined in this report. Inclusion of by-products may result in positive associative effects that are not properly accounted for in many nutrition models, such as the case when rumen pH is increased when starch is replaced with digestible fiber (Bradford and Mullins, 2012). This chapter groups each by-product feed into one of five categories and includes a brief description of feeds, including nutrient composition, availability, and impact on milk production.

POORLY DIGESTIBLE FIBER

Feeds in this class can be useful because they provide effective fiber that stimulates rumination and the formation of a rumen mat; they can be useful in diluting the diet concentrations of highly fermentable carbohydrates such as starch, which can reduce or modify the concentrations of organic acids produced in the rumen (Allen et al., 2009). Compared to forages, many of these feeds have a smaller particle size, and this may have a faster rumen passage rate and greater feed intake. When not used as feed, some products within this category are used in other applications on the dairy farm such as a source of bedding.

Corn cobs and Residue

The structural makeup of corn residue is about 21 percent cobs, 54 percent stalks, 22 percent leaves, and 13 percent husks (NRC, 1983). Corn cobs are high in fiber, low in protein, and poorly digested (Nangole et al., 1983), but compared to most by-products, the particle size of corn cobs is fairly coarse (Mertens, 1997). Partial replacement of alfalfa silage with corn cobs has been evaluated, and despite the fact that the concentration of energy in the diet was reduced, cows consumed more feed and milk yield was not negatively affected (Depies and Armentano, 1995). When used to replace corn, the addition of corn cobs negatively affected milk yield likely because of reduced supply of digestible energy (Soper et al., 1977). In general, crop residues such as corn stover are poorly digested but may be improved through alkaline treatment (Klopfenstein and Owen, 1981). Example of these treatments includes sodium hydroxide (NaOH), ammonia (NH₃), calcium hydroxide (Ca(OH)₂), potassium hydroxide (KOH), and calcium oxide (CaO) (Watson et al., 2015). Although safety concerns surround alkaline treatment, replacing wheat hay with corn residue treated with NaOH in rations fed to dairy cattle increased fiber digestibility and energy-corrected milk (Jami et al., 2014). The replacement of Chinese wild rye, corn silage, or corn grain with distillers grains and corn stover treated with CaO has shown promise to maintain production and reduce feed costs, but more research is needed, especially in high-producing cows (Shi et al., 2015).

Cottonseed Hulls

Cottonseed hulls are the outer covering of cottonseed (AAFCO, 2016) and a product of the mechanical removal of oil and meal from the cottonseed. Despite the low nutrient content, cottonseed hulls are highly palatable to cattle (Rogers et al., 2002). Cottonseed hulls may be used to partially replace forage fiber, but to maintain milk production, increased inclusion of energy sources such as corn grain is likely required (Shin et al., 2012). When cottonseed hulls were included at 8 percent of the diet dry matter (DM), rumination activities were reduced while feed intake was increased (Kononoff and Heinrichs, 2003). Cottonseed hulls may be useful when included in the starter diet of calves (Hopkins, 1997), but it should be noted that young animals may be particularly sensitive to gossypol. The addition of cottonseed hulls in a low-fiber calf starter mix increased feed intake, average daily gain, and postweaning body weight (Hill et al., 2009a). Although generally low in protein, the concentration may increase if greater internal portions of the seed itself are present.

Cotton Gin Trash

This is the lowest-value residue produced from the ginning of cotton and formally referred to as “cotton plant by-product” and contains cotton burrs (husks), leaves, stems, lint, immature

seeds, and/or dirt (AAFCO, 2016). Cotton gin trash is generally considered a poor-quality feed for dairy cattle, and the chemical composition is highly variable. Gin trash is high in lignin and, due to contamination of soil, is also high in ash. Cotton gin trash is also very coarse in texture and possess a low bulk density, making transportation difficult. Historically, there has been risk of cattle consuming this feed experiencing toxicity due to the insecticide known as disulfoton; however, use of this has been reduced (Rogers et al., 2002). When cotton gin trash replaced dehydrated alfalfa cubes, milk yield was reduced (Brown et al., 1979).

Oat Hulls

Oat hulls are a by-product of oat milling and are a high-fiber feed. This fiber is also highly lignified. Although the extent of fiber digested in the rumen is poor, the degree of cell wall lignification varies by genotype, and this affects the extent to which fiber is digested (Thompson et al., 2000). Methods of increasing the digestibility of fiber through chemical treatment such as alkaline hydrogen peroxide (Cameron et al., 1991a,b; Titgemeyer et al., 1991) have been evaluated, and when treated oat hulls were fed to cows in mid-lactation in place of alfalfa and corn silage, feed intake and production of fat-corrected milk increased (Cameron et al., 1991a). This method has not been widely adopted due to the caustic nature of the substance and associated safety risks (Shreck, 2013).

Peanut Hulls and Peanut Skins

Peanut by-products may contain mycotoxins, with aflatoxins being the most common. Producers should obtain an analysis of aflatoxin content to prevent possible aflatoxin poisoning of cattle and contamination of milk (Hill, 2002). Peanut hulls are not commonly fed to dairy cattle because the digestibility is extremely low (Huffman and Duncan, 1952). Ruminal DM digestibility of peanut hulls is 25 percent but may be increased to 40 percent through chemical treatment with NH₃ or NaOH (Barton et al., 1974). Peanut hulls are often ground and, as a result, low in effective fiber. Compared to hulls, peanut skins are higher in protein and fat while also lower in fiber. Peanut skins are also high in tannins, which may react with proteins and form protein-tannin complexes and reduce protein availability and may negatively affect palatability (West et al., 1993).

Pineapple Cannery Waste

In regions of the world where pineapples are grown, planting and harvesting of pineapple occur year round (Bartholomew et al., 2003). The nutrient content and in vitro digestibility of postharvest pineapple plant material, namely roots, stump, ratoon stems, green leaves, and dried leaves, have been evaluated (Kellems et al., 1979). Pineapple cannery waste is composed

of the outer peel (shell), crown and bud ends of the fruit, fruit trimmings, the inner core, and the pomace. The exact proportions of these parts and their associated chemical composition vary by variety and processing methods (Devendra, 1985). This feed may be fed either fresh or ensiled (Suksathit et al., 2011; Gowda et al., 2015), but the DM content is low, and if not stored correctly, it may spoil quickly (Nhan et al., 2009).

Rice Hulls

Rice hulls or husks consist of the outer covering of the rice grain and along with rice bran is a by-product of rice grain milling (Vadiveloo et al., 2009; AAFCO, 2016). This feed is low in protein and high in fiber and ash. It may be used as a low-quality animal feed but also as a fertilizer, an industrial energy source, or even as a filler for lignocellulosic fiber-thermoplastic composites (Vadiveloo et al., 2009). The digestibility of rice hulls is poor, and this is rarely fed to dairy cattle as a source of energy (Daniels and Hashim, 1977).

Sugarcane Bagasse, Silage, or Hay

This is a poor-quality roughage and is the pulp remaining from the removal of leaves and tops and the extraction of sugar from sugar cane (Fadel, 1999; AAFCO, 2016). This fibrous feed contains approximately 80 percent neutral detergent fiber (NDF), but the digestibility is poor because it is high in silica and lignin (Walford, 2008). Chemical treatment of bagasse increases its digestibility and milk production when fed (Randel et al., 1972).

Tomato Pomace

This by-product is produced during the production of tomato paste, juice, sauce, or ketchup and is usually composed of water, skins, seeds, and other hard tissues of the fruit (El Boushy and Poel, 1994). When produced, the moisture content is high, and as a result, it is often dried to aid in transportation and storage (Weiss et al., 1997). Tomato pomace is high in fiber but also contains approximately 5 percent pectin (Del Valle et al., 2007). A recent advancement in processing allows the separation of seeds from pulp and skin/peel with the seeds being produced as a feed by-product. Compared to tomato pomace, this feed is high in fat and protein and can replace whole cottonseed in rations fed to lactating cows without affecting milk production (Cassinero et al., 2015).

DIGESTIBLE ENERGY

This class of by-product feeds is fed to dairy cattle because they have moderate to high digestibility and can be used to replace either forages or grains. These by-products are low in starch, but they often contain sugars, digestible fiber, and soluble fiber that contribute energy to rumen microbes, which, in

turn, produce volatile fatty acids (FAs) and microbial protein (Dann et al., 2014). These products can replace higher-starch feeds. For example, concentration of starch was reduced from 27 to 18 percent with the addition of nonforage fiber sources, namely, beet pulp and brewers grains, without affecting the flow of microbial protein out of the rumen (Hristov and Ropp, 2003). The inclusion rate of these by-products can be exceptionally high without any negative effects on milk production if diets are formulated properly.

Almond Hulls

Almond hulls are a by-product from harvesting procedures for almond nuts (Fadel, 1999). Almond hulls are composed of primarily the mesocarp surrounding the fruit (Grasser et al., 1995). High in digestible fiber, this by-product is also high in fermentable sugars, which varies by variety (Offeman et al., 2014). Although a common feedstuff, few studies have evaluated its effect on milk production. As a partial replacement for forage, almond hulls have maintained milk production in one study (Aguilar et al., 1984) but not in another (Williams et al., 2018). The chemical composition of almond hulls has been reported to be affected not only by the variety of almond but also by the amount of debris present (DePeters et al., 2020). Like many by-products, proper dry storage is important to maintain the nutritional quality of this feedstuff as moisture may lead to mold growth and washout loss of sugars.

Beet Pulp

The production, chemical composition, and nutritional value of beet pulp have been reviewed (Kelly, 1983; Minnich et al., 2017). Beet pulp is the by-product when sugar is extracted from sugar beets (Fadel, 1999) and may be fed in wet, dry, pelleted, or ensiled forms and may also contain varying amounts of added molasses (Asadi, 2007). Beet pulp is high in NDF but also contains appreciable and variable concentrations of soluble fiber and sugars (DePeters et al., 2000). Beet pulp can be used as a replacement of high-starch grains in rations fed to dairy cattle. When beet pulp replaced high moisture corn and was included at up to 24 percent of the diet DM, milk production was maintained and rumen microbial nitrogen efficiency was not affected (Voelker and Allen, 2003a,b,c). The ruminal digestibility of fiber in beet pulp is rapid (DePeters et al., 1997) and is thought to at least be in part due to the higher arabinose content of hemicellulose. Adding molasses increases the sugar content and dilutes the concentration of fiber (Fadel et al., 2000). Sugar beet pulp silages in France were assayed for common mycotoxins, and although low concentrations were detected in 20 percent of the samples, the concentrations were not high enough to present a health risk for either animals or consumers (Boudra et al., 2015). Like many by-products, the chemical composition of beet pulp is known to vary among sources, and thus

chemical composition based on source may be useful for diet formulation procedures (Arosemena et al., 1995).

Citrus Pulp

Resulting from the extraction of juice, citrus pulp is made up of the ground peel, pulp, and seed residues of citrus fruit. Although this feed may contain any citrus crop, based on worldwide citrus production, oranges probably are the source of more than two-thirds of the citrus pulp produced (Crawshaw, 2004; Bampidis and Robinson, 2006). Citrus pulp is low in protein but supplies energy in the form of fermentable fiber, pectin, and sugars. The production and physical characteristics as well as the nutrient composition and value of citrus pulp have been reviewed (Bampidis and Robinson, 2006). Cattle consume this feedstuff in the wet form, but it is often dried to improve shelf-life and to enhance delivery logistics. When fresh citrus pulp is used, it should not be stored for long periods of time as remaining sugars will support secondary fermentation and mold growth and may attract insects (Bampidis and Robinson, 2006). Dehydrated citrus pulp containing mold can be the cause of citrinin toxicosis. Citrinin is produced by *Aspergillus* and *Penicillium* spp. (Gupta, 2007). During the dehydration process, CaO or Ca(OH)₂ may be added to the residue to release bound water (Arthington et al., 2002). Therefore, dried citrus pulp is usually high in Ca but low in phosphorus (P) and must be considered when formulating diets (Bath et al., 1980). A summary of studies in which citrus pulp was fed to lactating dairy cattle replacing corn grain or other high-starch ingredients concluded milk production and composition are usually maintained (Bampidis and Robinson, 2006). Although citrus pulp is commonly fed to dairy cattle with no ill effects, a delayed or type IV hypersensitivity reaction has occurred in a small number of cows consuming citrus pulp and is lethal (Saunders et al., 2000; Iizuka et al., 2005). Citrus pulp naturally contains phytochemicals, such as essential oils, which are antimicrobial, and feeding citrus pulp to ruminants has reduced both cecal and rectal populations of *Escherichia coli* 0157:H7 (Callaway et al., 2011). Citrus pulp may contain pesticide residues, but given the concentrations commonly detected and typical inclusion rates, milk safety concerns are not likely (Fink-Gremmels, 2012).

Crude Glycerol

A by-product of biodiesel production, as the name implies, this by-product is high in glycerin but also contains low concentrations of water, ash, trace minerals, free FAs, and methanol (Ma and Hanna, 1999). A portion of glycerol may escape rumen fermentation and be available as a glucogenic substrate (Werner Omazic et al., 2015). The use of glycerol in dairy rations has been reviewed (Donkin, 2008; Meral et al., 2015). This feed has been used to replace energy sources in the diets of lactating cows and has maintained (Donkin et al., 2009; Carvalho et al., 2011) or increased (Shin et al., 2012; Gaillard et al., 2018) milk yield.

Fruit and Vegetable By-Product

The chemical composition and nutritional characterization of fruit and vegetable waste have been reviewed (Angulo et al., 2012). This feed is usually used where the fruits and vegetables are grown, but occasionally it is transported further distances (Froetschel et al., 2014). Because this feed may contain almost any fruit or vegetable in either a whole or processed state, the chemical composition is highly variable. As a result, adequate sampling and analysis of this feed are important. Given the high moisture content, this feed is highly perishable but in some cases may be ensiled and then fed to dairy cattle (Yang et al., 2010; Kotsampasi et al., 2017). Several studies have attempted to characterize microbes, which may be present in this by-product, and although present, these studies did not detect levels that would be considered dangerous for livestock (Sancho et al., 2004; Angulo et al., 2012). Nonetheless, care should be taken to store it in conditions that will not encourage spoilage and contamination.

Potato Waste

Potato waste by-products can contain filter cake, steam peel, potato screenings, cull potatoes, and cooked or dried potato products (Crawshaw, 2004; Nelson, 2010). The different potato by-products, their associated chemical composition, and their use as a feed for cattle have been reviewed (Nelson, 2010). In addition, the use of the potato for feed has been reviewed (Whittemore, 1977). The chemical composition of these by-products varies widely. Consequently, it is important to adequately sample and analyze the product that is fed. The DM content of potato waste is usually low, and consequently, cost of transportation and storage may be a challenge. However, the high moisture and starch content often enable producers to ensile this feedstuff effectively. Compared to potato waste from processing, the DM and fat contents of dried potato products are higher (Rooke et al., 1997). The starch from potatoes is approximately 25 percent amylose and 75 percent amylopectin (French, 1973) and is generally less fermentable in the rumen than the starch found in most grains (Monteils et al., 2002; Mosavi et al., 2012). Potato waste may contain a number of antinutritional constituents or toxic substances. Sprouted and sunburned or green potatoes may contain toxic glycoalkaloids most commonly α -chaconine and α -solanine (see Chapter 17). Although pesticides are used in potato production, if used properly, the risk to animal health is minimal (Nelson, 2010). Although animals may learn how to safely chew whole or coarsely chopped potatoes, feeding these to cattle may result in choking and death (Bradshaw et al., 2002).

Soyhulls

Early in the oil recovery process, the hull or seed coat is removed from the soybean. The hull accounts for about 8 percent of bean DM. Historically, soyhulls were finely

ground and blended with the meal to obtain a crude protein (CP) content of 44 percent. This practice was a means of disposing of the hulls, which were of low value. Today, much of the soybean meal marketed does not contain soyhulls making them available for cattle fed. Soyhulls are often ground because it increases bulk density, aiding in pelleting and transportation (Anderson et al., 1988). Soyhulls are often heated to inactivate antinutritional factors (Johnson et al., 2008). The characteristics and nutritional value of soyhulls have been reviewed (Ipharraguerre and Clark, 2003). Although the fiber may be extensively digested by rumen microbes, *in vivo* digestion is lower than that *in vitro* or *in situ* digestibility (Ipharraguerre and Clark, 2003). This discrepancy may be in part due to the fine particle size of soyhulls, making them pass out of the rumen rapidly and as a consequence limit the extent of rumen digestion (Drackley, 2000). The concentration of CP is about 10 percent and can contribute appreciable amounts of rumen-degradable protein. Soybean hulls are generally included in the rations of lactating dairy cattle as a source of digestible energy and may replace either forages or concentrates (Firkins and Estridge, 1992; Ipharraguerre and Clark, 2003; Ranathunga et al., 2010). Replacing corn grain with soyhulls can have positive effects on rumen fermentation because they contain little if any starch, which helps maintain rumen pH. When corn grain in a diet was reduced from 40 to 1 percent by replacing it with soyhulls, yield of milk and milk components was maintained (Ipharraguerre et al., 2002a,b). As much as 30 percent of the diet DM may be soyhulls without negatively affecting ruminal fermentation, diet digestibility, or production of dairy cows.

Wheat Middlings

A by-product of the wheat milling process, wheat middlings are composed of wheat bran, shorts, germ, flour, and other assorted portions from the tail of the mill such as red dog (AAFCO, 2016). Shorts are mostly made up of fine bran particles while red dog consists mostly of the aleurone layer with small particles of bran, germ, and flour (Blasi et al., 1998). The chemical composition of wheat middlings has been reviewed (Boros et al., 2004; Slominski et al., 2004; Rosenfelder et al., 2013). This by-product is considered an energy feed because it is higher in starch and fiber but moderate in protein. It is also a good source of P (Erickson et al., 1985). The protein contained in this feed is highly degradable in the rumen (Batajoo and Shaver, 1998). Wheat middlings are commonly used to replace high-starch grains such as corn (Acedo et al., 1987; Bernard, 1997; Dann et al., 2014) but may also be used to replace some forage fiber (Wagner et al., 1993), but the small particle size reduces the effectiveness of fiber (Depies and Armentano, 1995). The chemical composition of wheat middlings may vary and is influenced by wheat type and variety as well as growing conditions, grade of flour produced, and the proportion of bran included (Blasi et al., 1998; Rosenfelder et al., 2013). Although preparatory

processes carried out before milling reduce the mycotoxin content of the grain, mycotoxins in these by-products may be up to 8-fold higher than the flour (Cheli et al., 2013).

Wheat Bran

Wheat bran originates from the wheat milling process and is composed of the pericarp and outer seed tissues, including the aleurone layer, while also containing varying amounts of endosperm (Rosenfelder et al., 2013; AAFCO, 2016). Compared to middlings, the feeding of wheat bran to dairy cattle is less common (Ertl et al., 2016), but it has greater nutritional value than rice bran (Tahir et al., 2002).

PROTEIN FEEDS

This class of by-product feeds is fed to dairy cattle because they contain high concentrations of protein. The extent to which protein is digested in the rumen and in the small intestine varies by feedstuff (Paz et al., 2014).

Animal Products

The use and safety of animal feed ingredients has been reviewed (Clark et al., 1987; Sapkota et al., 2007; Jayatilakan et al., 2012). About 30 to 50 percent of each animal used in the production of food is not consumed by humans. Instead, it goes through the rendering process in which it is exposed to heat while moisture is extracted and fat is separated. The resulting feed by-products include meat and bone meal, meat meal, poultry meal, hydrolyzed feather meal, blood meal, and animal fats (Meeker, 2006). Compared to oilseed meals, animal by-products supply more essential amino acids (AAs), especially lysine (Lys; Ravindran and Blair, 1993). The 2003 discovery of the first U.S. case of bovine spongiform encephalopathy (BSE) and concerns for bacterial contamination of animal feed on human bacterial illnesses have brought about establishment of new adjustments to existing restrictions on the use of many of these products (Garcia et al., 2006; Sapkota et al., 2007). In an attempt to prevent the spread of transmissible spongiform encephalopathy in the United States, the U.S. Food and Drug Administration (FDA, 2008) prohibits this use of specific cattle origin materials from the feed of all animals (see Chapter 17 for details). Regulations can change with time, and users of animal products for feed must be aware of and follow all current regulations regarding their use.

Blood Meal

Blood meal is high in CP, rumen-undegradable protein (RUP), and Lys (Boucher et al., 2009b). Subsequent to the slaughter of cattle, pigs, or poultry, blood is collected while avoiding contamination of hair, ingesta, or urine and processed to produce blood meal. The aim of blood processing is to obtain a dry, stable, nutrient-rich product that can be

ground to an even particle size. Blood meal is hygroscopic and must be processed to less than 12 percent DM and stored in dry facilities (Leoci, 2014). Blood is dried using several methods; the process begins with a coagulation process followed by batch drying, flash drying, or spray drying (Almeida et al., 2013). Very little blood meal is produced using batch-drying techniques, which essentially involve removal of water by cooking, steaming, or hydrolyzation. In flash drying, moisture in blood is first removed using a mechanical dewatering process, and it may be further condensed by cooking. The remaining semisolid material then undergoes rapid drying using a drum or ring drying process (AAFCO, 2016). In drum drying, blood is applied as a thin layer onto the outer surface of revolving horizontal drums that are internally heated by steam. After the product is dried, it is removed from the drum with a scraper and the dried blood is ground into flakes or powder (Tang et al., 2003). In ring drying, blood is simultaneously ground and dispersed into a high-velocity air stream (Pearson and Dutson, 1992). Whole blood or separated plasma and red albumin may be dried through the spray-dried process (Almeida et al., 2013). In spray drying, moisture is removed using low temperatures and evaporators under vacuum until the material is approximately 30 percent solids, after which it is further dried by spraying with a draft of warm dry air (AAFCO, 2016). The method of drying affects the availability of protein and AAs in this by-product (Batterham et al., 1986; Messman and Weiss, 1994).

Fish Meal

Fish meal is high in undegradable protein and methionine (Met; Chalupa and Sniffen, 1996; Santos et al., 1998); however, the consistency of the rumen-undegradable portion of protein can vary greatly across sources (Boucher et al., 2009a). The intestinal digestibility of RUP of this feed is high and the Na content may vary (Taghizadeh et al., 2005). Fish meal is also a good source of essential FAs (Ravindran and Blair, 1993). Fish meal can be made of clean, dried, ground tissue of whole fish and/or cuttings (AAFCO, 2016), which are then cooked, pressed, dried, and ground. Resulting fish meal may originate from a variety of different types of fish, including anchovy, herring, menhaden, pilchards, sardines, sharks, grayfish, catfish, and pollock (Ockerman and Hansen, 2000). The production process of fish meal has been reviewed (Ghaly et al., 2013), as has the use of fish meal in ruminant diets (Hussein and Jordan, 1991). The type of fish and drying temperatures can affect its chemical composition, the ruminal disappearance, and the intestinal digestibility (Opstvedt et al., 1984; Gonzalez et al., 1998; Boucher et al., 2009b). There are essentially two different methods used in the production of fish meal. In the wet process, which is most common, oil is removed while it is not removed in dry processing (Pearson and Dutson, 1992). Replacing blood meal with fish meal can have positive effects on milk production (Moussavi et al., 2007) but not always (Mattos et al., 2002). Replacing soybean meal with

fish meal often increases the yield of milk protein, a response likely due to increasing the supply of limiting AAs such as Lys or Met (Polan et al., 1997). Feeding of fish meal can increase the intake of eicosapentaenoic acid and docosahexaenoic acid, which may have positive effects on cow fertility (Burke et al., 1997; Mattos et al., 2002; Staples et al., 2005). These FAs will also be found in the milk, but the observed differences are not directly proportional to intake (Wright et al., 2003) because of extensive rumen biohydrogenation and preferential deposition into body tissue (AbuGhazaleh et al., 2004). No effects of feeding fish meal were observed on the physical, chemical, sensory, and processing properties of milk. However, fat globule size was smaller, churning time of cream was longer, and butter possessed a soft texture—effects all likely a result of changes in the FA composition of milk fat (Avramis et al., 2003). The inclusion of fish meal may negatively affect palatability of a total mixed ration (Shaver, 2005). This by-product is used in pet foods and in aquaculture, and as a result, less is available for livestock feed.

Feather Meal

Approximately 5 to 7 percent of the total body weight of domestic fowl is composed of feathers (Onifade et al., 1998). The production processes of feather meal have been reviewed (Papadopoulos, 1985; El Boushy et al., 1990). Using steam and pressure, feather meal is usually hydrolyzed to disrupt the structure of keratin to increase digestibility and to produce a sterilized product (Blasi et al., 1991; El Boushy and van der Poel, 1994). During this process, sulfur (S) is volatilized, and as a result, the S content of feather meal may be used to evaluate the extent of hydrolysis (Moritz and Latshaw, 2001). In some cases, both acidic and enzymatic additives may be used to reduce odor and improve nutritive value (El Boushy et al., 1990). Although high in CP and sulfur AAs, feather meal is low in histidine (His), Lys, and Met (Goedeken et al., 1990; Klemesrud et al., 2000; Stahel et al., 2014). Combining both feather meal and blood meal is an effective way to increase RUP and essential AA content (Grant and Haddad, 1998; Bargo et al., 2001). The combination of adding feather meal to bone meal to increase the supply of cysteine (Cys) to spare Met did not meet the Met needs of lactating cows (Pruekvimolphan and Grummer, 2001).

Meat and Bone Meal, Porcine

A rendered product made up of porcine tissues, meat and bone meal does not contain added blood, hair, hoof, hide, manure, or digesta (AAFCO, 2016). This by-product is high in RUP and essential AAs. Intestinal digestibility of RUP is also high (Howie et al., 1996; Klemesrud et al., 1997). When compared to all animal protein sources, this by-product usually has the lowest concentration of protein but is high in Ca and P because of the varying presence of bone (Ravindran and Blair, 1993; Traylor et al., 2005). The nutritional quality

and sources of variation of meat and bone meal have been reviewed (Hendriks et al., 2002, 2004). When replacing soybean meal, meat and bone meal can increase efficiency of protein utilization, likely due to the high concentration of RUP, high intestinal digestibility of this feed, and increasing the supply of essential AAs (Akayezu et al., 1997).

Poultry By-Product Meal

Poultry by-product meal consists of ground, rendered carcass parts and, except what may be unavoidable, does not contain feathers (AAFCO, 2016). This by-product is extensively used by the pet food industry, and as a result, less is available for livestock feed (Dozier and Dale, 2005). Its chemical composition has been reviewed (Dale et al., 1993; Dozier and Dale, 2005). In addition to a high protein content, this feed is high in fat, P, Ca, and essential AAs. It can be fed to cattle and used as a source of RUP (Bohnert et al., 1998, 1999) and essential AAs (Mantysaari et al., 1989; Polan et al., 1997).

Whey

The composition of whey and several whey components as well as responses to animals consuming these products has been reviewed (Schingoethe, 1976; Kosikowski, 1979). Whey is high in moisture, lactose, and Na. Despite the fact that casein is removed during the cheese-making process, it contains moderate concentrations of protein. Like all dairy products, whey is also a good source of Ca and P. Whey is the principal by-product of the dairy processing industry. Sweet whey is a by-product of cheese production while acid whey is a by-product of the production of acidified products, namely, cottage cheese, yogurt, or casein. Given the diversity of milk processing techniques and products produced, variation in chemical composition of by-products designated as whey or whey products is high. The increased use of microfiltration and production of Greek yogurt has increased the production of acid whey. Acid whey is lower in protein and higher in minerals, notably Ca, than sweet whey (Smith et al., 2016). This difference in composition has made the development of methods to produce dry whey ingredients from Greek yogurt problematic. Therefore, the liquid products are available for animal feed (Lagrange et al., 2015). Animals consuming whey may exhibit increased urination, likely a response to the high concentration of Na. Animals may also scour while the erosion of teeth has also been observed. Whey may also be deproteinized, resulting in the by-product called whey permeate, which is much lower in protein but is still high in lactose, minerals, and moisture.

Canola Meal

Canola is a trademarked name for rapeseed, which contains <2 percent erucic acid in the oil and <30 pmol alkenyl glucosinolates per gram of oil-free DM. This is reduced from 25 to 45 percent erucic acid and 50 to 100 pmol glucosinolates

in conventional rapeseed meal (Bell, 1993). Glucosinolates are bitter, they impair palatability, and interfere with the synthesis of thyroid hormones by impairing the uptake of iodine (Woyengo et al., 2016). Canola meal is the meal remaining after the extraction of oil from *Brassica campestris*, *Brassica napus*, or *Brassica juncea* seeds by either mechanical or solvent extraction methods (AAFCO, 2016). The chemical composition, ruminal degradability, and lactational performances of dairy cows consuming canola meal have been reviewed (Mustafa et al., 2000; Huhtanen et al., 2011; Martineau et al., 2013).

Compared to soybean meal, canola meal contains less protein and energy, but the protein is less degradable in the rumen. In a meta-analysis, Huhtanen et al. (2011) found that canola meal was at least as good as soybean meal and that some improved responses are due to increases in feed intake. A more recent meta-analysis conducted by Martineau et al. (2013) evaluated the replacement of other sources of protein with canola meal. In general, milk and milk protein yields increased with canola meal. Canola meal contains goitrogenic compounds that can reduce the transfer of iodine into milk (Laarveld et al., 1981; Weiss et al., 2015). See Chapter 7 for more details. Bell (1993) outlined other constituents of canola meal that may exert antinutritional effects. Canola meal contains sinapine, which may negatively affect palatability, but this has not been reported in cattle. Sinapine affects the flavor of eggs from chickens, but no research has been conducted on potential effects on milk flavor. Canola meal contains approximately 1.5 to 3.0 percent tannins, which may interfere with protein digestion.

Cottonseed Meal

This by-product is produced when the oil from cottonseed has been removed. Oil may be removed either through mechanical or solvent extraction, and the resulting meal must not contain less than 36 percent CP (AAFCO, 2016). The protein content and quality of cottonseed meal are high, and when replacing either soybean meal or canola meal, little difference in milk production is usually observed (Brito and Broderick, 2007).

In a number of studies, milk protein decreased when cows consume cottonseed meal (Maesoomi et al., 2006; Brito and Broderick, 2007). This may be because of the lower concentration of Lys. In addition, the bioavailability of Lys in cottonseed meal may be low because under heat, gossypol binds to proteins, notably the epsilon amino group of Lys (Reiser and Fu, 1962; Blauwiekel et al., 1997).

Soybean Meal

Soybean meal products are discussed in Chapter 19.

Sunflower Meal

A by-product of the process of extracting oil from sunflower seed, sunflower meal has moderate concentrations of

TABLE 15-1 Major Coproducts Resulting from the Dry Grain Corn-Milling Process

Name	Brief Description
Condensed distillers solubles	Resulting from the removal of ethyl alcohol by the distillation from the yeast fermentation of grain by condensing this stillage to a semisolid state.
Com bran	Produced when the com ethanol facility dehulls and degerms the grain subsequent to fermentation. This feed has not gone through the fermentation process.
Com germ	Produced when the com ethanol facility dehulls and degerms the grain subsequent to fermentation. This feed has not gone through the fermentation process. Because the germ contains much of the fat found in the com kernel, in addition to fiber, the product is also high in fat.
Deoiled distillers dried grains with solubles	Resulting product from solvent extraction of oil from com distillers dried grains with solubles (DDGS), and fat content may be as low as 3 percent on an as-fed basis. Term solvent extracted is not required by AAFCO (2016).
Distillers dried grains with solubles	Resulting from the removal of ethyl alcohol by distillation from the yeast fermentation of grain or a grain mixture by condensing and drying at least 75 percent of the solids of the whole stillage.
Distillers dried solubles	Resulting from the removal of ethyl alcohol by distillation from the yeast fermentation of grain or a grain mixture by separating the coarse grain fraction of the whole stillage and drying it.
High-protein distillers grains	Grains after the com ethanol facility dehulls and degerms the grain and then is fermented. In most cases, solubles are not added back as traditionally done in the dry milling process, but when this does occur, this product is called high-protein distillers with solubles. The protein content of this feed is similar to soybean meal and contains less phosphorus than traditional DDGS.
Modified wet distillers grains with solubles	Partially dried and contain approximately 50 to 54 percent moisture (Mello et al., 2012).
Reduced-fat distillers dried grains with solubles	Produced when the solubles are centrifuged before they are added to the distillers grains.
Wet distillers grains	Resulting from the removal of ethyl alcohol by the distillation from yeast fermentation of grains.
Wet distillers grains with solubles	Resulting from the removal of ethyl alcohol by distillation from the yeast fermentation of grain or a grain mixture by condensing and drying at least 75 percent of the solids of the whole stillage.

protein and fiber (Arieli et al., 1999). In general, it is a good source of digestible protein, but the fiber is highly resistant to rumen digestion (Van Soest, 1994). The chemical composition of sunflower meal has been reviewed (Lardy and Anderson, 2002; Lomascolo et al., 2012; Itavo et al., 2015). Sunflower meal is high in copper and has been documented to contribute to copper toxicity in sheep, which are particularly sensitive to this mineral (Garcla-Fernandez et al., 1999). This feed is also high in S containing AAs, such as Met, but compared to soybean meal is low in Lys. The chemical composition of sunflower meal varies depending on the method of processing. Solvent extraction methods are most efficient at removing oil, resulting in sunflower seed with a low fat content, but mechanical expeller or extrusion methods also exist (Drackley and Schingoethe, 1986; Lardy and Anderson, 2002). The extent of hull removal affects the chemical composition. Sunflower meal probably contains more degradable protein than soybean meal or canola meal (Lardy and Anderson, 2002). Sunflower meal has been used effectively as a replacement for soybean meal without affecting the yield or composition of milk (Schingoethe et al., 1977). Sunflower meal is also used in calf starter diets (Drackley and Schingoethe, 1986) and was more palatable than canola meal (Miller-Cushon et al., 2014).

Safflower Meal

Compared to soybean or canola, safflower is a minor oilseed crop (Ekin, 2005). A by-product of the process in

which oil is extracted out of the safflower seed, safflower meal is good source of both protein and fiber. The crude fat content varies depending on the extraction process but is lowest when solvent extraction is used. The protein appears to be relatively resistant to degradation by rumen microbes (Dixon et al., 2003a,b). The feed contains two phenolic glucosides, namely, matairesinol- β -glucoside and the purgative 2-hydroxyarctiin- β -glucoside that may make the feed bitter and unpalatable to cattle (Jin et al., 2010). Glucosinolates have been reviewed (Tripathi and Mishra, 2007).

ENERGY/PROTEIN BY-PRODUCTS

Corn Ethanol Production, the Dry Grind Process

Over 30 percent of the corn crop in the United States is probably used to produce fuel ethanol. Since approximately 2004, the fuel ethanol industry has experienced dramatic growth. A number of reviews have been published outlining the chemical composition and use of corn milling coproducts in rations for dairy cattle (Schingoethe et al., 2009; Hollmann et al., 2011a,b; Bradford and Mullins, 2012; Paz et al., 2014; Bottger and Sudekum, 2018). The growth of the dry milling industry has brought about increased availability of corn milling coproducts (see Table 15-1). The dry milling process (see Figure 15-1) is used at approximately 80 percent of the corn ethanol facilities in the United States, and it begins when the entire corn kernel is ground through a hammer mill

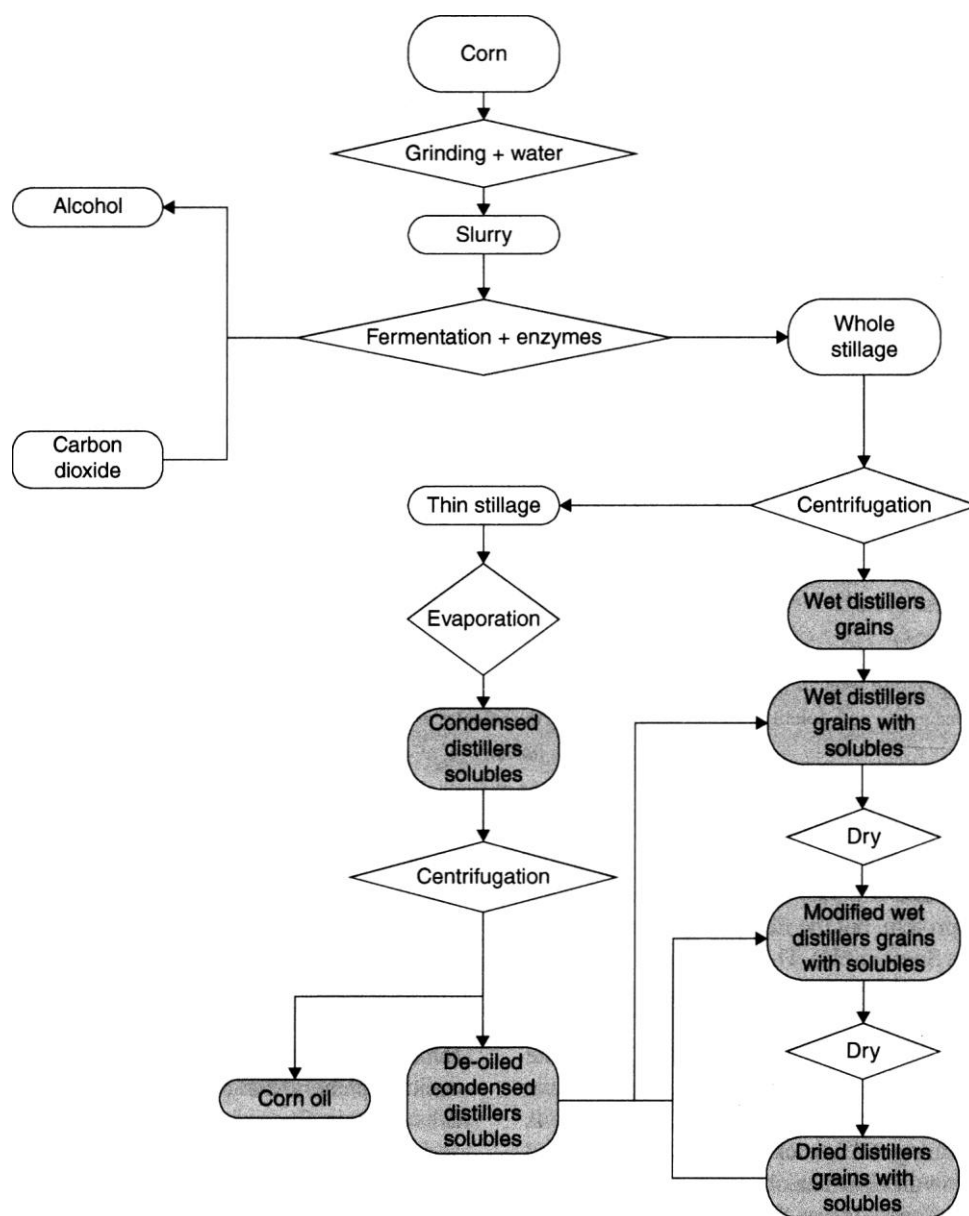


FIGURE 15-1 Schematic representation of the dry milling ethanol process for corn ethanol production. Diamonds denote a processing step and ovals denote a product. Feed coproducts are shaded gray.

and hot water is added (Rosentrater, 2015). After about 50 to 60 hours of fermentation, the resulting mash contains about 15 percent ethanol (Bothast and Schlicher, 2005). The ethanol is distilled off, and the remaining residue is called whole stillage and contains yeast cells, nonfermented solids, and water. The stillage can be centrifuged, resulting in “thin stillage,” which contains 5 to 10 percent solids, and wet distillers grains (WDG). The thin stillage can be evaporated, producing syrup that can be blended with WDG to produce WDG with solubles that are often dried to produce dried distillers grains with solubles (DDGS). Compared to corn grain, protein, fat, and most minerals increase about 3-fold, but concentrations of Na, S, and Ca may be greater because exogenous sources

of these minerals may be added during the production process (NRC, 2012).

Corn milling coproducts may contain mycotoxin that were present in the incoming grain. If they withstand the fermentation process, the concentrations of these toxins can increase 3-fold in the final feedstock because a large proportion of starch is removed during the process (Fink-Gremmels, 2012). The toxins can include aflatoxins, deoxynivalenol, fumonisins, ergot alkaloids, and zearalenone. A 2-year (2006 to 2008) study evaluating 235 DDGS samples collected from 20 U.S. corn ethanol plants and 23 export shipping containers found that none of the samples contained aflatoxins or deoxynivalenol concentrations greater than FDA guidelines, and no more

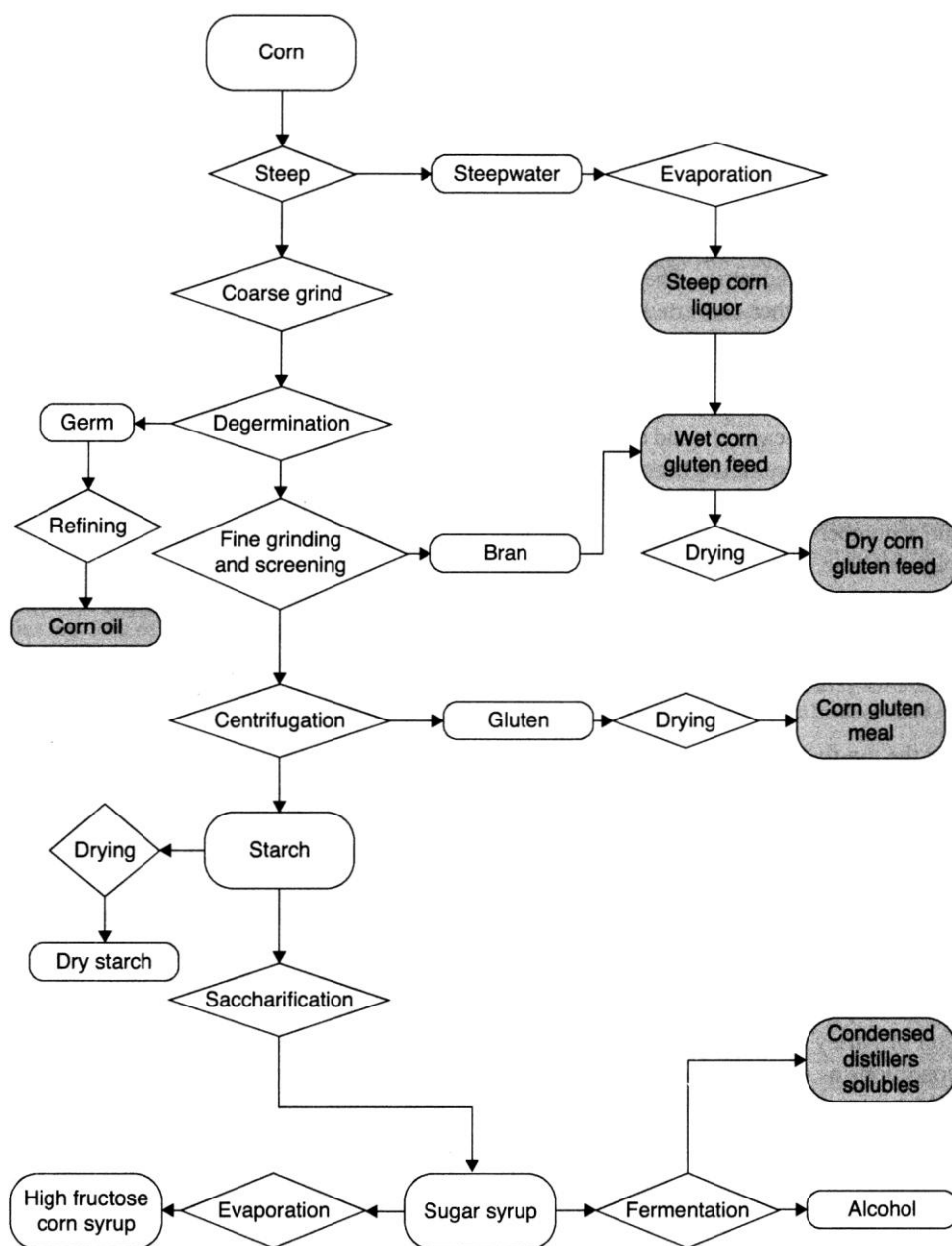


FIGURE 15-2 Schematic representation of the wet milling process for corn ethanol and high-fructose corn syrup. Diamonds denote a processing step while the ovals are products; feed coproducts are shaded gray.

than 10 percent of the samples contained fumonisin at concentrations higher than the recommendation for feeding equids and rabbits (Zhang et al., 2009). Fumonisin was detected at concentrations lower than FDA guidelines for use in animal feed. In addition, none of the samples contained T-2 toxins higher than the detection limit, while zearalenone concentrations were lower than the detection limit in most samples.

The Wet Milling Process

Compared to the dry grind process, the wet milling process (see Figure 15-2) is more energy and capital intensive (Bothast

and Schlicher, 2005). After cleaning, the corn is steeped in a dilute solution of sulfurous dioxide for approximately 40 hours to start breaking down the protein in the grain. The resulting steepwater can be condensed, resulting in a product that may be sold as steep corn liquor (referred to by AAFCO [2016] as condensed fermented corn extractives) or used as a source of nutrients for subsequent fermentation. This liquor is composed of a mixture of soluble components of corn grain containing both protein and complex sugars (Crawshaw, 2004). At this point, the corn kernel goes through a series of grinds, differential separations, and centrifuges. The germ is isolated, from which the oil is extracted. The remaining coproduct at this stage is corn *germ*

meal (wet milled) (Crawshaw, 2004; AAFCO, 2016). Once the germ is removed, the residue is milled to release the starch and aid in removal of the hull or bran. The bran is then pressed, and usually the com steep liquor is added back to produce com gluten feed. Com gluten feed may be sold in either wet or dry forms. With much of the fiber removed, the remaining material is centrifuged to separate the gluten and starch fractions. The remaining com gluten meal is high in zein protein and usually is used in the pet food and poultry industries. The starch from the wet milling process can be further converted to dextrose and then fermented to produce fuel ethanol or used in other industrial fermentation processes. If used to produce ethanol, a resulting coproduct is distillers solubles. The solubles produced by the wet milling industry contain yeast cells and unfermented sugars and protein, but because the germ has been removed, it does not contain high concentrations of fat. The starch may be further processed into a number of food-grade products such as com syrup and high-fructose com sweetener (Stock et al., 2000). Com germ from the wet milling process has been fed to dairy cattle (Miller et al., 2009).

Brewers Grains

Beer ranks among the top five most consumed beverages in the world (Fillaudeau et al., 2006), and its by-product is brewers grains, or brewers spent grains. Brewers grains are mostly made up of the husk-pericarp-seed coat layers of the barley grain while hop residue will also be present (Mussatto et al., 2006; McCarthy et al., 2013). Brewers grains may also include com, wheat, rice, and other common sources of carbohydrates. Chemical variations of this by-product exist both across and within a brewery (Westendorf et al., 2014) and are largely a function of barley variety, harvest time, mashing, and mashing conditions as well as the quality and type of added adjuncts (Mussatto et al., 2006). Brewers grains contain cellulose, hemicellulose, lignin, protein, vitamins, and minerals (Priest and Stewart, 2006). The phenolic component is believed to have bioactive antioxidant potential but has not been studied in ruminants (McCarthy et al., 2013). As with other by-products from the food industry, fresh brewers grains are almost always a safe and nutrient-dense feed for dairy cows. The high moisture content of wet brewers grains makes it an unstable feedstuff, but it can be preserved by reducing the moisture content by pressing and then drying at the production facility (Aliyu and Bala, 2013). Brewers yeast is a by-product of the brewing industry and can be fed to cattle (Grieve, 1979). Brewers yeast is distinguished from yeast culture and yeast extract and other yeast-containing feeds (AOAC, 2016). Yeast used by brewers was selected or developed based on their effects on beer flavor or to optimize beer-making fermentation. Yeast used as direct feed additives was selected or developed for effects in the rumen (see Chapter 16). Usually available in a wet form, brewers yeast may be dried (Steckley et al., 1979a,b). When wet brewers grains were added to the rations of lactating dairy

cattle at 9 and 17 percent of the diet DM in replacement of forage, no effects on milk production were observed (Firkins et al., 2002). The RUP content of wet brewers grains is usually lower than that of WDG because RUP is lower in barley (brewers grains) than it is for corn (distillers grains). In addition, brewers grains are not exposed to heat during a distillation process. Drying brewers grain increases the RUP content because of heat exposure.

Bakery Waste

The chemical composition of bakery waste can be highly variable (Waldroup et al., 1982; Slominski et al., 2004) and originates from bread, cereal, or cookie production. The feed composition database of this publication distinguishes these sources. All types are high in starch while cookie waste is usually high in crude fat and sucrose or other simple sugars. In general, bakery waste contains a high concentration of energy and may be used as a partial replacement for cereal grains (Humer et al., 2018).

Cottonseed, Whole

The nutrient composition and impact on animal performance of cottonseed and cottonseed meal has been reviewed (Coppock et al., 1987; Arieli, 1998). There are two types of cottonseed available in the United States: (1) upland cotton (*Gossypium hirsutum*) or high lint and (2) Pima cotton (*Cosmosipium barbadense*), which is delinted. Pima cottonseed makes up only about 5.5 percent of the U.S. cotton production and is higher in fat, protein, and gossypol (Broderick et al., 2013). This by-product is usually fed after it has been cracked. Upland cottonseed is usually fed whole and is higher in fiber and should be stored in a facility where it is protected from moisture and well ventilated to prevent the formation of condensation and mold. Cottonseed should be tested for gossypol and mycotoxins. Details on both those toxins are in Chapter 17. Whole cottonseed is an excellent source of effective fiber for dairy cattle and often increases milk fat when fed (Clark and Armentano, 1993). The seed coat at least partially protects the oil from rumen microbes and biohydrogenation (Palmquist and Jenkins, 1980). Partial replacement (15 percent of diet DM) of both corn silage and alfalfa silage with whole cottonseed did not affect milk production and composition (Firkins et al., 2002). Cottonseed is an excellent source of protein and can effectively replace soybean meal in the ration of lactating dairy cows (Broderick et al., 2013).

High-Fat By-

Products Animal

Fats

Animal fats are a by-product of the meat industry, and in the United States, these most commonly originate from beef or pork processing. In the rendering process, heat and pres-

sure are used to separate lipid material from meat tissues. The American Fats and Oils Association (Columbia, SC) outlines marketing grades for animal fats that are based on characteristics such as melting point, color, density, moisture, and impurities (USDA, 2013). These grades include tallow, choice white grease, and yellow grease. These specifications are not based on the species of origin but rather outline the technical specifications above (Pearson and Dutson, 1992). Commercially rendered fat from cattle is often referred to as tallow, but it is technically defined as fat possessing titer temperature (temperature at which FAs of a given fat solidify) greater than 40°C. Rendered fat from swine is usually commercially referred to as lard or grease, but technically, lard and grease have a titer equal to or less than 40°C (Ockerman and Hansen, 2000). The nutritional quality of different animal fats is dependent on the FA composition, which is a function of the animal species from which they originate. The FA profile of the diet fed to swine has a major impact on the FA profile of rendered pork fat. Because of rumen biohydrogenation by rumen microbes, diet has less of an impact on the FA profile of rendered beef fat. Compared to beef fat, pork fat is softer, and this is due to the higher concentration of linoleic acid. Pork fat is also lower in myristic acid (unless pigs are fed beef fat) and is also essentially void of trans unsaturated FAs (Berger, 1997). Yellow grease originates from restaurant cooking practices but may also originate from rendering plants producing lower-quality greases. The impact of animal fats on milk production is related to the associated effects on intake, rumen fermentation, and digestibility of the FAs (see Chapter 4). Animal fats are also added to milk replacers (Jenkins et al., 1986; Hill et al., 2009b) to increase energy density. Animal fats may be added to calf starters (Hill et al., 2015) to increase energy density but are more frequently used to control dust. Because animal fat sources are susceptible to oxidation (Shurson et al., 2015; Joseph, 2016), antioxidants are often added (Buck, 1991).

Rice Bran

Rice bran is largely made up of the pericarp and germ of the rice grain but may also contain other constituents of the rice plant. In some cases, a portion of the oil is removed through the use of solvents (AAFCO, 2016). Both conventional (Nornberg et al., 2004; Wang et al., 2015; Criscioni and Fernandez, 2016) and defatted (Chaudhary et al., 2001) rice bran can be fed successfully to cattle.

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